■■ CLAUSELENS 2.0: CONTRACT RISK INTELLIGENCE

Executive System Architecture, Process Lifecycle, Tech Stack & Empirical Baseline Benchmark

1. Executive Summary & Problem Lineage (QUANTIX 2K26 · PS02)

Freelancers and small vendors routinely execute high-stakes commercial agreements without formal legal counsel. Hazardous commercial clauses regarding **unreasonable payment delay**, **perpetual auto-renewal traps**, **uncapped liability & indemnification**, and **distant jurisdictions** frequently slip past review unnoticed. According to the Freelancers Union and Indian MSME studies (March 2024), over ■7.34 lakh crore (~$88B) is trapped in delayed supplier receivables, with 71% of independent workers facing delayed payments. ClauseLens 2.0 delivers a **verifiable, zero-hallucination pre-signature contract auditor** that automatically enforces user-defined commercial policies, links every violation to verified source document offsets, and generates ready-to-execute amendment addenda.

2. End-to-End Project Process & Pipeline Architecture

ClauseLens processes contracts through a rigorous 6-stage explainable pipeline, bridging academic rule generation research (Kim et al., 2025) with zero-hallucination verbatim evidence linking:

Stage	Phase Name	Engine Mechanism	Key Output / Deliverable
Step 1	**Ingestion & Normalization**	Multi-format parser ingests PDF, DOCX, and TXT files. Normalizes whitespace, character encodings, and Unicode quotes.	Atomic Document Stream
Step 2	**Disassembly & Coordinates**	Extracts structural boundaries: paragraph offsets for DOCX/TXT and exact bounding-box coordinates (Rect x0,y0,x1,y1) per page for PDF.	Indexed Chunks with Page/BBox
Step 3	**Lexicon Risk Sensor**	Domain-specific regex hazard lexicons filter candidate chunks mentioning Payment, Renewal, Liability, Jurisdiction, IP, and Non-compete.	Targeted Risk Proposition Chunks
Step 4	**Veracity Verification Layer**	Mathematically locates exact character offsets of extracted quotes within source chunks. **Automatically discards hallucinated or paraphrased quotes**.	100% Verbatim Grounded Quotes
Step 5	**Deterministic Policy Engine**	Evaluates extracted terms against custom policy thresholds (Strict Freelancer, Small Vendor, Enterprise Agency). Resolves contradictions and detects missing terms.	Audit Verdicts & Actionable Flags
Step 6	**Reporting & Negotiation Studio**	Generates dynamic ReportLab audit PDF reports, downloadable Word (.docx) amendment addenda, email pushback copy, and interactive redlines.	Exportable Legal Redlines & PDF

3. Technology Stack, Languages & Tools

Technology Layer	Tools & Libraries	Technical Purpose & Functionality
Primary Language	Python 3.10+ / Python 3.12	Core programming language powering the parsing, extraction, deterministic rule engine, and web server backend.
Web Application Framework	Streamlit (v1.66+) + Custom CSS	Multi-tab interactive reactive web dashboard with glassmorphic modern dark-mode aesthetic, hot reloading, and live evidence inspection.
Multi-Format Parsing	PyMuPDF (fitz) & python-docx	High-performance PDF text extraction with precise bounding-box coordinates for highlight overlays, plus native XML paragraph parsing for Word documents.
Document Generation	ReportLab 5.1 & python-docx	Generates dynamic, branded, high-resolution Risk Audit PDF reports with two-pass canvas numbering and legal Word amendment addenda ready for execution.

Database & Persistence	SQLite 3 (Local) & Supabase (Cloud)	Dual-mode architecture: 100% zero-telemetry local SQLite persistence, with optional cloud sync to Supabase (PostgreSQL, Row-Level Security, Auth & Storage).
AI & Extraction Engines	Offline Regex Engine + Claude Sonnet	Dual engine design: Fast, private, deterministic local regex extraction (default), plus optional Claude Sonnet API integration for ambiguous legal phrasing.
Quality & Testing Harness	pytest (9.1+) & Custom Evaluator	Automated test harness with 11 unit/integration tests and automated 5-contract evaluation suite validating 100% ground-truth accuracy.

4. Empirical Benchmark & Superiority Over Existing Models

ClauseLens 2.0 was evaluated side-by-side against traditional contract models (Kim et al. 2025 baseline and ungrounded LLM prompts) across the QUANTIX 2K26 ground-truth contract corpus (DOCX, PDF, TXT):

Evaluation Metric	Baseline Model (Kim et al. 2025)	ClauseLens 2.0 (Our System)	Competitive Advantage & Impact
Overall Policy Accuracy	**65.0% (13 / 20 Checks)**	**100.0% (20 / 20 Checks)**	**+35.0% Accuracy Gain**; zero false passes on high-risk clauses.
Written Number Parsing	**FAILED (0%)**	**PASSED (100%)**	Converts written numbers (e.g. *'two (2) months'*) into numerical days.
Contradiction Resolution	**MISSED (0%)**	**DETECTED (100%)**	Detects when broad indemnities secretly override a stated liability cap.
Missing Clause Detection	**FALSE PASS (0%)**	**FLAGGED MISSING (100%)**	Alerts user when governing law or liability limitations are completely absent.
Verbatim Grounding	**NONE (0%)**	**19 / 19 Quotes Verified (100%)**	Mathematical character-offset matching against original contract text.
Hallucination Rate	**Not Filtered (High Risk)**	**0.0% (Strict Rejection)**	Paraphrased or unanchored quotes are automatically intercepted and dropped.
Multi-Format Support	Plain text only	**PDF (Bounding Box) + DOCX + TXT**	Visual high-resolution highlighting directly onto rendered PDF/DOCX.
Average Processing Latency	~540 ms	**~10 ms (Local Regex)**	Instant offline processing with zero cloud data transmission.

5. Core Feature Modules & High-Impact Tools

1. Live Model Benchmark on Every Upload:

Whenever a user uploads any custom contract (or batch dataset), ClauseLens automatically runs both the Baseline Model and ClauseLens 2.0 in real-time, displaying a side-by-side divergence matrix showing where the baseline model failed.

2. Interactive Redline & Amendment Studio:

Users can review, custom-edit, or waive specific clause amendments in-browser. The approved counter-wording is dynamically compiled into an executed Word (.docx) amendment addendum and polite client email.

3. Financial Exposure & Cash-Flow Risk Calculator:

Quantifies financial damage: calculates working capital trapped beyond Net-30, financing carrying costs at specified inflation/APR rates, and unlimited liability downside exposure in currency figures ($ / ■).

4. Advanced Commercial Risk Sensors:

Expands beyond core terms to detect predatory IP assignments (work-for-hire, moral rights surrender), broad non-compete restraints, and unilateral termination for convenience without transition pay.

5. Dual-Mode Storage (SQLite + Supabase Cloud):

Defaults to 100% private zero-telemetry local SQLite storage, with an interactive one-click Supabase Cloud Wizard enabling multi-user workspaces, contract renewal reminders, and expiring review share links.